
Parametric Density Estimation II

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Outline

- Density Estimation
- Maximum Likelihood vs Bayesian Learning
- Maximum Likelihood Estimation
- Maximum A-Posteriori (MAP) Estimation
- Example: The Gaussian Case
- Bayesian Learning
- Recursive Bayesian Incremental Learning

Parametric Learning

- Bayesian Decision

$$\arg \max_i p(\omega_i | x)$$

- Maximum Likelihood Estimation

$$\arg \max_{\theta_i} p(D | \theta_i)$$

- Maximum A-Posteriori Estimation

$$\arg \max_{\theta_i} p(\theta_i | D), \theta_i = \text{constant}$$

- Bayesian Learning (Estimation)

Find $p(\theta_i | D), \theta_i = \text{random variable}$

Example of Maximum Likelihood

The Gaussian multivariate case

- For the multivariate case,

$$\ln p(x_k | \mu) = -\frac{1}{2} (x_k - \mu)^t \Sigma^{-1} (x_k - \mu) - \frac{d}{2} \ln 2\pi - \frac{1}{2} \ln |\Sigma|$$

- MLE estimates are given by

$$\hat{\mu} = \frac{1}{n} \sum_{k=1}^n x_k$$

$$\hat{\Sigma} = \frac{1}{n} \sum_{k=1}^n (x_k - \hat{\mu})(x_k - \hat{\mu})^t$$

- The MLE for Σ is **biased** (i.e., the expected value over all data sets of size n of the sample variance is not equal to the true variance, that is,

$$E \left[\frac{1}{n} \sum_{k=1}^n (x_k - \hat{\mu})(x_k - \hat{\mu})^t \right] = \frac{n-p}{n} \Sigma \neq \Sigma$$

Example of Maximum Likelihood:

The Gaussian multivariate case

- **Unbiased estimator** for μ and Σ are given by

$$\hat{\mu} = \frac{1}{n} \sum_{k=1}^n x_k$$

$$\hat{\Sigma} = \frac{1}{n-p} \sum_{k=1}^n (x_k - \hat{\mu})(x_k - \hat{\mu})^t$$

- $\hat{\Sigma}$ is called the **unbiased sample covariance matrix**

$$E \left[\frac{1}{n-p} \sum_{k=1}^n (x_k - \hat{\mu})(x_k - \hat{\mu})^t \right] = \frac{n-p}{n-p} \Sigma = \Sigma$$

- **Biased** sample variance $\hat{\Sigma}$ is *asymptotically unbiased*.

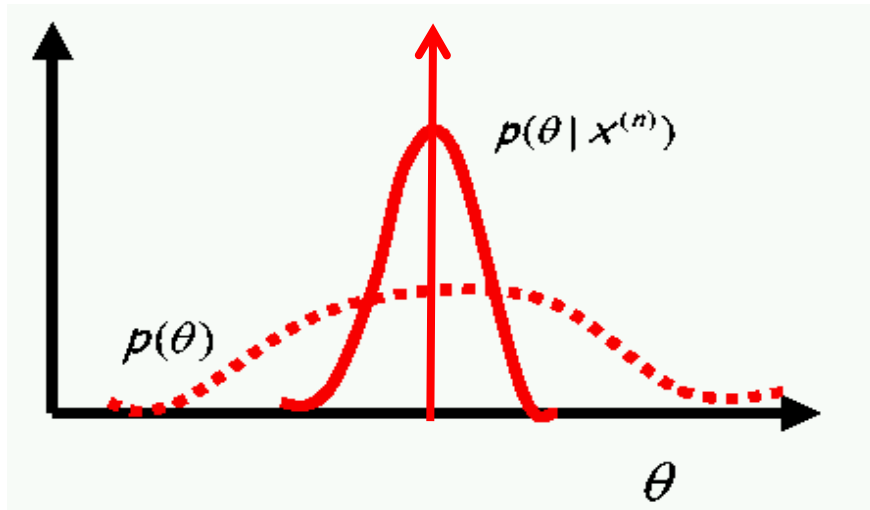
$$E \left[\frac{1}{n} \sum_{k=1}^n (x_k - \hat{\mu})(x_k - \hat{\mu})^t \right] = \frac{n-p}{n} \Sigma \rightarrow \Sigma \quad \text{as } n \rightarrow \infty$$

Bayesian Learning

- The aim of decision is to find posteriors $p(\omega_i|x)$ knowing $p(x|\omega_i)$ and $p(\omega_i)$, but they are unknown.
- How to estimate them from the training sample set D ?
- Generally used assumptions:
 - ✓ Priors generally are known or obtainable from a trivial calculations. Thus $p(\omega_i) \sim p(\omega_i|S)$. cf. $p(\omega_i|S) \neq p(\omega_i|x)$. S includes label.
 - ✓ The training set can be separated into c subsets: $D_1, \dots, D_c$
 - ✓ D_j have no influence on $p(\mathbf{x}|\omega_i, D_i)$ if $i \neq j$.
 - ✓ Let the estimate of $p(\mathbf{x}|\omega_i)$ be $p(\mathbf{x}|\omega_i, D_i)$ for each class
$$p(\mathbf{x}|\omega_i) = p(\mathbf{x}|\omega_i, D_i) \cong p(\mathbf{x}|D_i) := p(\mathbf{x}|\theta_i)$$
 - ✓ Bayesian learning: find $p(\theta_i|D_i) \xrightarrow{\text{omit index}} p(\theta|D)$

Bayesian Learning: General Theory

- *Bayesian learning* considers $\theta \leftarrow \theta_i \leftarrow \omega_i$ (the parameter vector to be estimated) to be a **random variable**.
- Before we observe the data, the parameters are described by a **prior** $p(\theta)$ which is typically very broad. Once we observed the data, we can make use of Bayes' formula to find **posterior** $p(\theta|D)$. If the values of parameters are more consistent with the data, the **posterior is narrower than prior**. This is *Bayesian learning* (see fig.)



$p(\mathbf{x}|\omega_i) = p(\mathbf{x}|\omega_i, D_i) \cong p(\mathbf{x}|D_i) \leftarrow p(\mathbf{x}|\theta), p(\theta|D)$
where θ is latent random variable.

Find $p(\theta|D)$ by Bayes rule

$$p(\theta|D) \propto p(D|\theta) p(\theta)$$

Bayesian Learning: General Theory

- Density function for x , given the training data set D ,

$$p(x|\omega) \cong p(x|D) = \int p(x, \theta|D) d\theta = \int p(x|\theta, D) p(\theta|D) d\theta$$

- By independence between x and D , $p(x|\theta, D) = p(x|\theta)$
- Therefore $p(x|D) = \int p(x|\theta) p(\theta|D) d\theta$ ($= p(x|\theta)$ if $p(\theta|D)$ is impulse).
- Instead of choosing a specific value for θ , the Bayesian approach performs a weighted average over all values of $p(\theta)$
- The weighting factor $p(\theta|D)$, which is a posterior of θ , is determined by starting from some assumed prior $p(\theta)$

Bayesian Learning: General Theory

- Then update it using Bayes' formula to take account of data set D . Since $D = \{x_1, \dots, x_n\}$ are drawn independently,

$$p(D|\theta) = \prod_{k=1}^n p(x_k|\theta),$$

which is likelihood function.

- Posterior for θ is

$$p(\theta|D) = \frac{p(D|\theta)p(\theta)}{p(D)} = \frac{p(\theta)}{p(D)} \prod_{k=1}^n p(x_k|\theta),$$

where normalization factor $p(D) = \int p(\theta) \prod_{k=1}^n p(x_k|\theta) d\theta$.

Bayesian Learning

– Univariate Normal Distribution

- Let us use the Bayesian estimation technique to calculate a *posteriori* density $p(\theta|D)$ and the desired probability density (likelihood) $p(x|D)$ for the case $p(x|\mu) \sim N(\mu, \Sigma)$, where rv θ for the parameter μ .
- Univariate Case: $p(\mu|D)$
- Let μ *be the only unknown* parameter

$$p(\mathbf{x}|\omega_i) \sim p(x|\mu) \sim N(\mu, \sigma^2)$$

Bayesian Learning

– Univariate Normal Distribution

- **Conjugate** prior probability: **normal** distribution over μ ,

$$p(\mu) \sim N(\mu_0, \sigma_0^2)$$

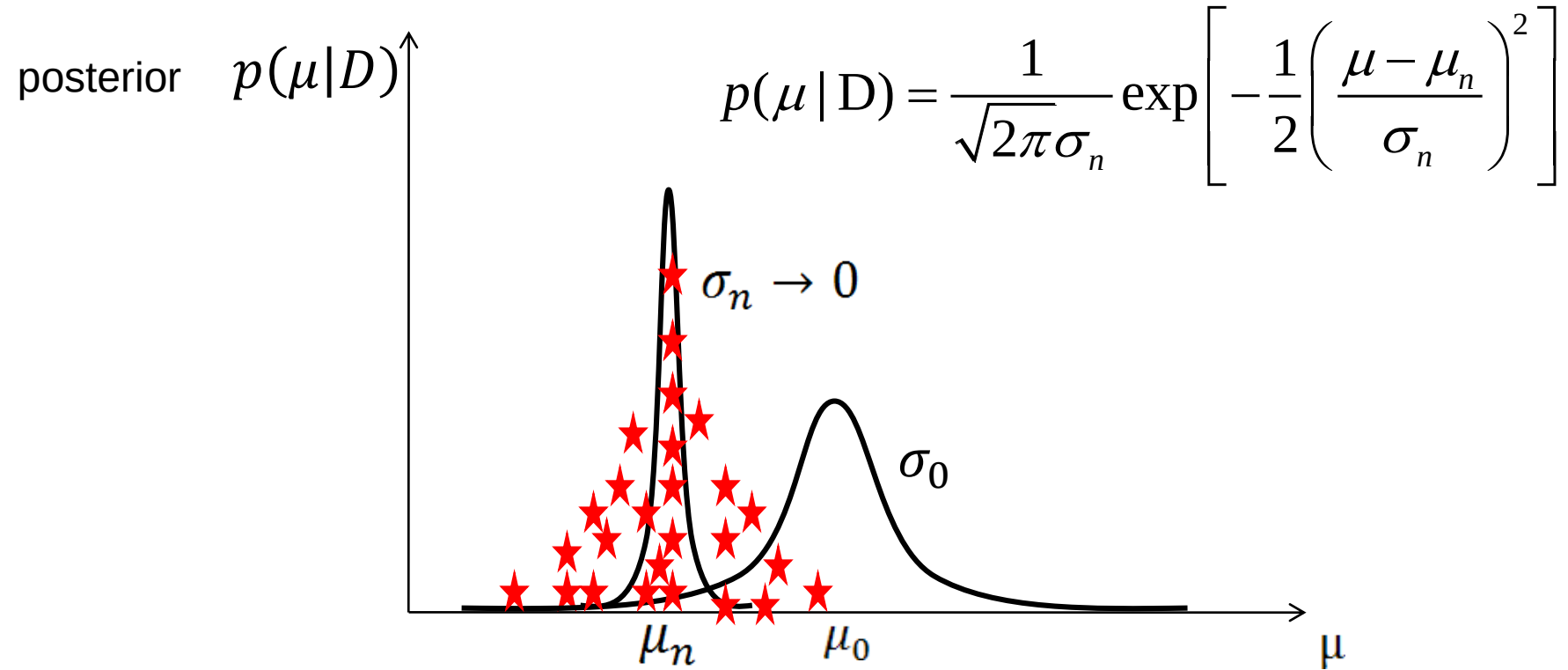
encodes some prior knowledge about the true mean μ_0 , while σ_0^2 measures our prior uncertainty.

- If μ is drawn from $p(\mu)$ then density for x is completely determined. Letting $D = \{x_1, \dots, x_n\}$, we use

$$\begin{aligned} p(\mu|D) &= \frac{p(D|\mu)p(\mu)}{\int p(D|\mu)p(\mu)d\mu} \\ &= \alpha \prod_{k=1}^n p(x_k|\mu)p(\mu). \end{aligned}$$

Bayesian Learning

– Univariate Normal Distribution



Bayesian Learning

– Univariate Normal Distribution

$$p(x|\mu) \sim N(\mu, \sigma^2)$$

$$p(\mu) \sim N(\mu_0, \sigma_0^2)$$

- Computing the posterior distribution

$$p(\mu|D) = \alpha p(D|\mu)p(\mu)$$

$$= \alpha' \exp \left[-\frac{1}{2} \left(\sum_{k=1}^n \left(\frac{x_k - \mu}{\sigma} \right)^2 + \left(\frac{\mu - \mu_0}{\sigma_0} \right)^2 \right) \right]$$

$$= \alpha'' \exp \left[-\frac{1}{2} \left(\left(\frac{n}{\sigma^2} + \frac{1}{\sigma_0^2} \right) \mu^2 - 2 \left(\frac{1}{\sigma^2} \sum_{k=1}^n x_k + \frac{\mu_0}{\sigma_0^2} \right) \mu \right) \right]$$

- Since $p(\mu|D)$ should be Gaussian

$$p(\mu|D) = \frac{1}{\sqrt{2\pi}\sigma_n} \exp \left[-\frac{1}{2} \left(\frac{\mu - \mu_n}{\sigma_n} \right)^2 \right] = \alpha''' \exp \left[-\frac{1}{2\sigma_n^2} \mu^2 - \frac{\mu_n}{\sigma_n} \mu \right]$$

- $\mu_n = f(x_k, \mu_0, \sigma_0) = ?$, $\sigma_n = h(x_k, \mu_0, \sigma_0) = ?$

Bayesian Learning

– Univariate Normal Distribution

- Solution: $\frac{1}{\sigma_n^2} = \frac{n}{\sigma^2} + \frac{1}{\sigma_0^2}$, $\frac{\mu_n}{\sigma_n^2} = \frac{n}{\sigma^2} \hat{\mu}_n + \frac{\mu_0}{\sigma_0^2}$

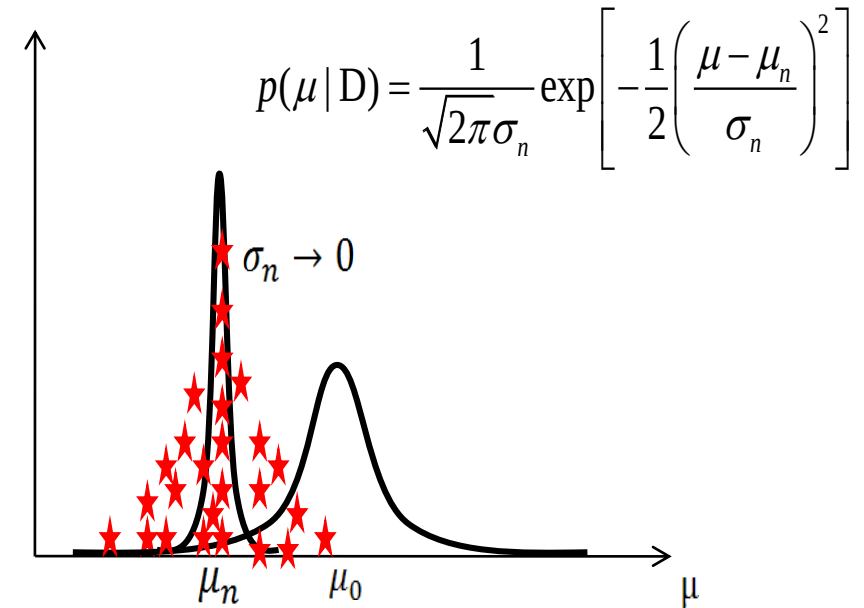
where $\hat{\mu}_n = \frac{1}{n} \sum_{k=1}^n x_k$ is the sample mean.

- Solving explicitly for μ_n and σ_n^2 we obtain

$$\sigma_n^2 = \frac{\sigma_0^2 \sigma^2}{n\sigma_0^2 + \sigma^2}, \quad \mu_n = \left(\frac{n\sigma_0^2}{n\sigma_0^2 + \sigma^2} \right) \hat{\mu}_n + \frac{\sigma^2}{n\sigma_0^2 + \sigma^2} \mu_0$$

- μ_n represents our best guess for μ after observing n samples.
- σ_n^2 measures our uncertainty about this guess.
- σ_n^2 decreases monotonically with n

(approaching $\sigma_n^2 \rightarrow \frac{\sigma^2}{n} \rightarrow 0$ as n approaches infinity)

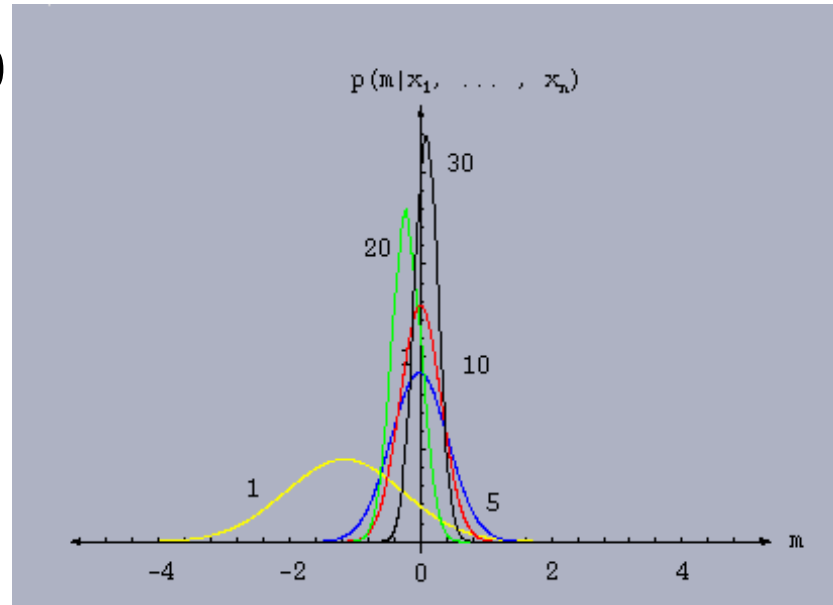


Bayesian Learning

– Univariate Normal Distribution

- Each additional observation decreases our uncertainty about the true value of μ .
- As n increases, μ becomes more and more sharply peaked, approaching a Dirac delta function as n approaches infinity. This behavior is known as *Bayesian Learning*.

$$p(\mu | D)$$



Bayesian Learning

– Univariate Normal Distribution

$$\mu_n = \left(\frac{n\sigma_0^2}{n\sigma_0^2 + \sigma^2} \right) \hat{\mu}_n + \frac{\sigma^2}{n\sigma_0^2 + \sigma^2} \mu_0, \quad \sigma_n^2 = \frac{\sigma_0^2 \sigma^2}{n\sigma_0^2 + \sigma^2}$$

- In general, μ_n is a linear combination of $\hat{\mu}_n$ and μ_0 , with coefficients that are non-negative and sum to 1.
 $\mu = \mu_0$
- Thus μ_n lies somewhere between $\hat{\mu}_n$ and μ_0 .
- If $\sigma_0 \neq 0$, $\mu_n \rightarrow \hat{\mu}_n$ as $n \rightarrow \infty$.
- If $\sigma_0 = 0$, our a priori certainty that $\mu = \mu_0 = \mu_n$ is so strong that no number of observations can change our opinion.
- If $\sigma_0 \gg \sigma$, a priori guess is uncertain, and we take $\mu_n = \hat{\mu}_n$
- The ratio $\frac{\sigma^2}{\sigma_0^2}$ is called *dogmatism*.

Bayesian Learning

– Univariate Normal Distribution

- The Univariate Case:

$$\begin{aligned} p(x|D) &= \int p(x|\mu)p(\mu|D)d\mu \\ &= \int \frac{1}{2\pi\sigma\sigma_n} \exp \left[-\frac{1}{2} \left(\frac{(x-\mu)}{\sigma} \right)^2 - \frac{1}{2} \left(\frac{(\mu-\mu_n)}{\sigma_n} \right)^2 \right] d\mu \\ &= \frac{1}{2\pi\sigma\sigma_n} \exp \left[-\frac{1}{2} \frac{(x-\mu_n)^2}{\sigma^2 + \sigma_n^2} \right] f(\sigma, \sigma_n) \\ &\propto \exp \left[-\frac{1}{2} \frac{(x-\mu_n)^2}{\sigma^2 + \sigma_n^2} \right] \xrightarrow{n \rightarrow \infty} \exp \left[-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2} \right] \end{aligned}$$

Bayesian Learning

– Univariate Normal Distribution

- Since $p(x|D) \propto \exp \left[-\frac{1}{2} \frac{(x-\mu_n)^2}{\sigma^2 + \sigma_n^2} \right]$, we can write

$$p(x|D) \sim N(\mu_n, \sigma^2 + \sigma_n^2)$$

- To obtain the class conditional probability $p(x|D)$, whose parametric form is known to be $p(x|\mu) \sim N(\mu, \sigma^2)$.
- We replace μ by μ_n and σ^2 by $\sigma^2 + \sigma_n^2$.
- The conditional mean μ_n is treated as if it were the true mean, and the known variance is **increased** by σ_n^2 to account for the additional uncertainty in x resulting from our **lack** of exact knowledge of the mean μ .

Recursive Bayesian Incremental Learning

- We have seen that $p(D^n|\theta) = \prod_{k=1}^n p(x_k|\theta) = p(x_n|\theta) \prod_{k=1}^{n-1} p(x_k|\theta)$. Let us define $D^n = \{x_1, \dots, x_n\}$. Then

$$p(D^n|\theta) = p(x_n|\theta)p(D^{n-1}|\theta).$$

- Substituting into $p(\theta|D^n)$ and using Bayes we have:

$$p(\theta|D^n) \propto p(D^n|\theta)p(\theta) = p(x_n|\theta)\boxed{p(D^{n-1}|\theta)}p(\theta)$$

$$= p(x_n|\theta)\boxed{\frac{p(\theta|D^{n-1})p(D^{n-1})p(\theta)}{p(\theta)}}$$

$$\propto p(x_n|\theta)p(\theta|D^{n-1})$$

Recursive Bayesian Incremental Learning

- While $p(\theta|D^0) = p(\theta)$, the repeated use of
$$p(\theta|D^n) \propto p(x_n|\theta)p(\theta|D^{n-1})$$

produces a sequence

$$p(\theta), p(\theta|x_1) = p(\theta|D^1), p(\theta|x_1, x_2) = p(\theta|D^2), \dots$$

- This is called the *recursive Bayes* approach to the parameter estimation. (Also *incremental* or *on-line* learning).
- When this sequence of densities **converges to a Dirac delta** function centered about the true parameter value, we have *Bayesian learning*.

Recursive Bayesian Incremental Learning

- Example of Recursive Formula

$$p(\theta|D^n) \propto p(D^n|\theta)p(\theta|D^0)$$

$$\theta_n = \left(\frac{n\sigma_0^2}{n\sigma_0^2 + \sigma^2} \right) \hat{x}_n + \frac{\sigma^2}{n\sigma_0^2 + \sigma^2} \theta_0$$

$$\sigma_n^2 = \frac{\sigma_0^2 \sigma^2}{n\sigma_0^2 + \sigma^2}$$

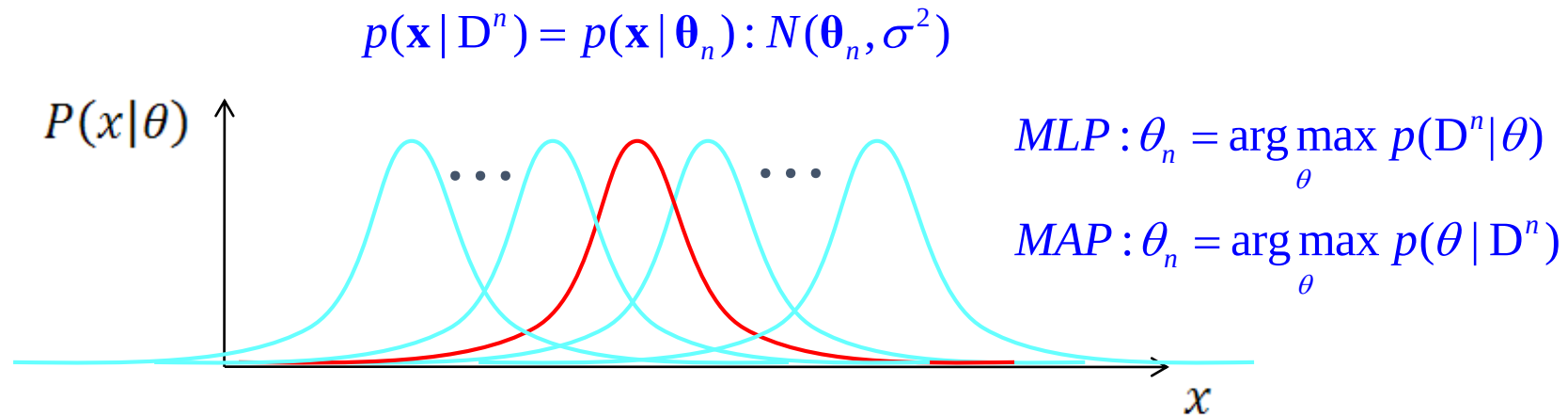
$$p(\theta|D^n) \propto p(x_n|\theta)p(\theta|D^{n-1})$$

$$\theta_n = \left(\frac{\sigma_{n-1}^2}{\sigma_{n-1}^2 + \sigma^2} \right) x_n + \frac{\sigma^2}{\sigma_{n-1}^2 + \sigma^2} \theta_{n-1}$$

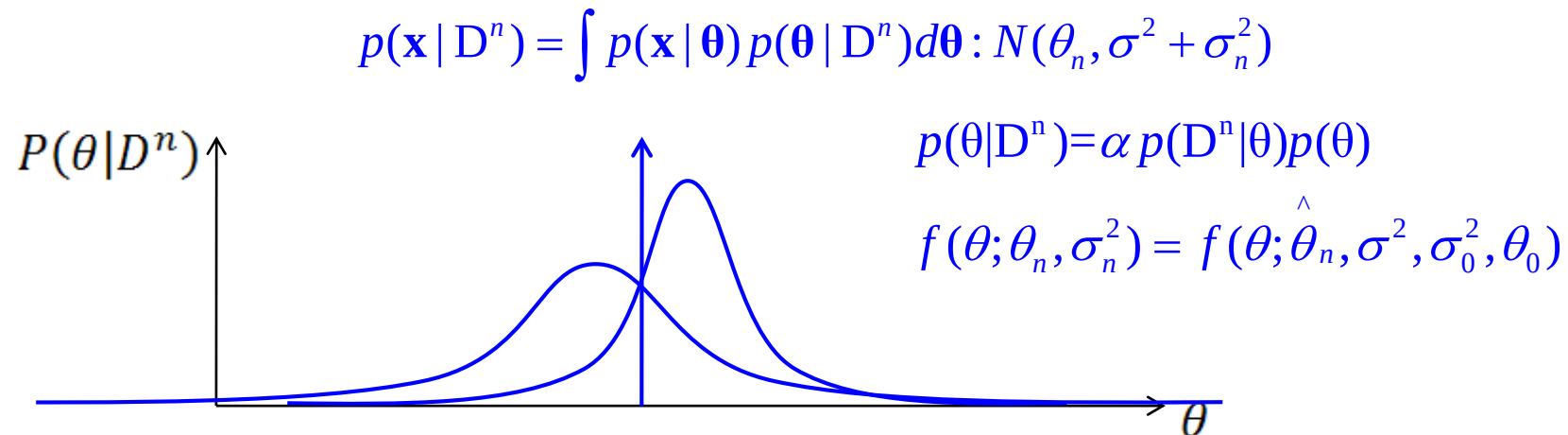
$$\sigma_n^2 = \frac{\sigma_{n-1}^2 \sigma^2}{\sigma_{n-1}^2 + \sigma^2}$$

Maximal Likelihood vs. Bayesian

- MLP, MAP



- Bayesian Learning



Maximal Likelihood vs. Bayesian

- ML and Bayesian estimations are asymptotically equivalent and “consistent”. They yield the same class-conditional densities when the size of the training data grows to infinity.
- ML is typically computationally easier: in ML we need to do (multidimensional) differentiation and in Bayesian (multidimensional) integration.
- ML is often easier to interpret: it returns the single best model (parameter) whereas Bayesian gives a weighted average of models.
- But for a finite training data (and given a reliable prior) Bayesian is more accurate (uses more of the information).
- Bayesian with “flat” prior is essentially ML.

Interim Summary

- In MLE estimation, $\hat{\mu} = \frac{1}{n} \sum_{k=1}^n x_k$, $\hat{\Sigma} = \frac{1}{n-p} \sum_{k=1}^n (x_k - \hat{\mu})(x_k - \hat{\mu})^t$
- In Bayesian Learning

- Weighted average of Candidate Models

$$p(x|D) = \int p(x|\theta)p(\theta|D)d\theta$$

- Gaussian Model

$$p(\theta|D^n) \propto p(D^n|\theta)p(\theta|D^0)$$

$$\theta_n = \left(\frac{n\sigma_0^2}{n\sigma_0^2 + \sigma^2} \right) \hat{\mu}_n + \frac{\sigma^2}{n\sigma_0^2 + \sigma^2} \theta_0$$

$$\sigma_n^2 = \frac{\sigma_0^2 \sigma^2}{n\sigma_0^2 + \sigma^2}$$

- Incremental Bayesian Learning

$$p(\theta|D^n) \propto p(x_n|\theta)p(\theta|D^{n-1})$$

$$\theta_n = \left(\frac{\sigma_{n-1}^2}{\sigma_{n-1}^2 + \sigma^2} \right) x_n + \frac{\sigma^2}{\sigma_{n-1}^2 + \sigma^2} \theta_{n-1}$$

$$\sigma_n^2 = \frac{\sigma_{n-1}^2 \sigma^2}{\sigma_{n-1}^2 + \sigma^2}$$